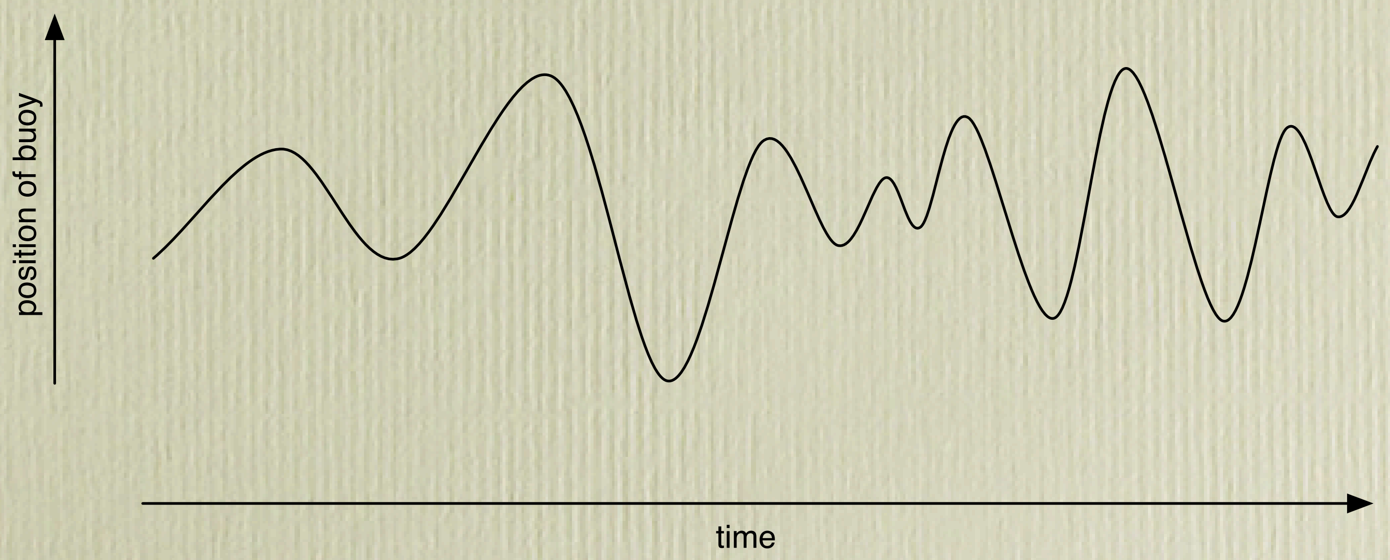
Spacetime diagrams

February 10th, 2004

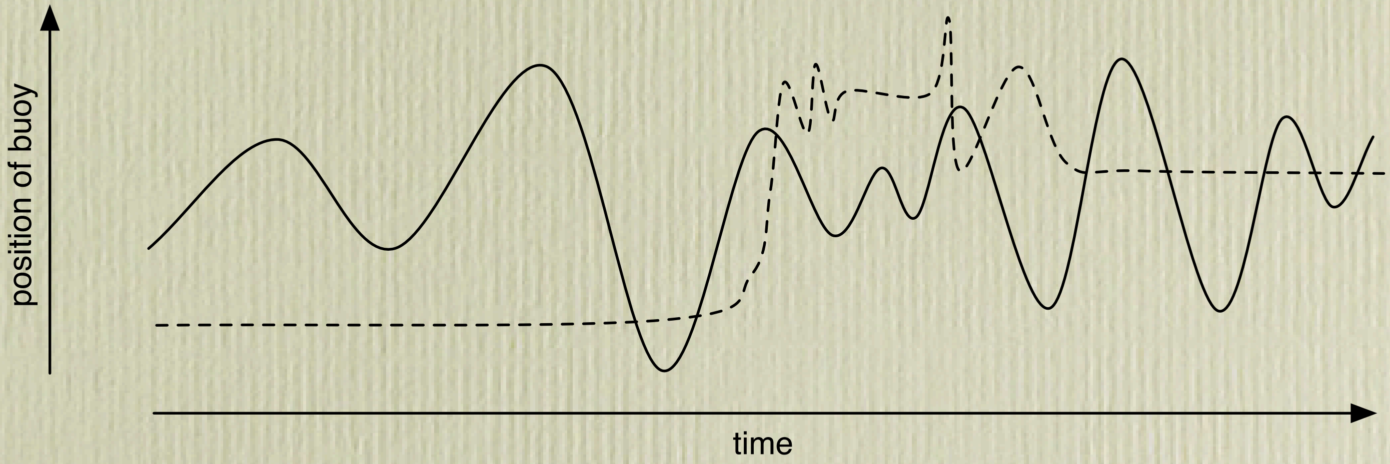
Maps of space and maps of time

- Maps of space are familiar.
- So are maps of time: they're called *timelines* and you can find them in children's history books.
- A timeline is one-dimensional. But you can use other dimensions of the map to convey information about quantities that vary over time.

- For example, information about the spatial position of something.
- A map showing how the height of a buoy attached to a post in the sea might vary over time:

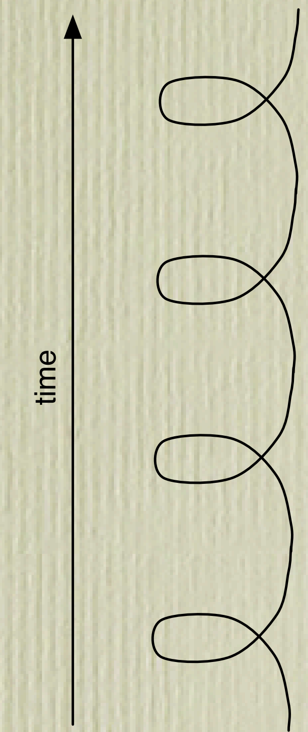


- A map showing the position both of the buoy and of a crab who moves up and down the post.



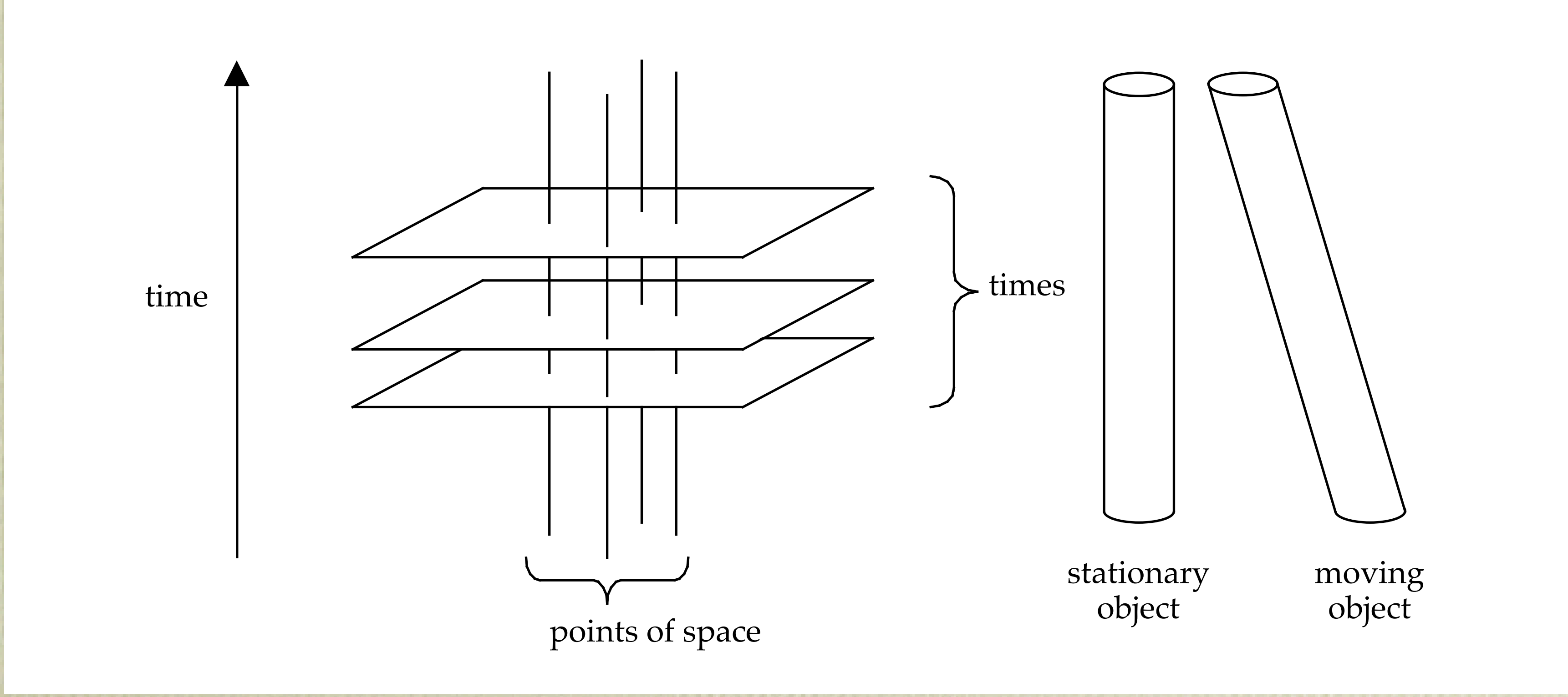
- The buoy and the crab are confined to a single spatial dimension. By constructing three-dimensional diagrams, we can chart the motions of objects that can move in two dimensions. In practice, we can use “fake 3d”.

- This diagram shows an object moving around in a circle, like a planet:



  - To depict objects moving in three dimensions, you'd need a four-dimensional diagram. So when we're making these diagrams we pretend that space is one- or two-dimensional.

- The 'Newtonian' way to read a spacetime diagram: each point in the diagram picks out (i) a point of space, and (ii) a time. A line parallel to the time axis corresponds to a point of space; a plane perpendicular to the time axis corresponds to a time. A stationary object traces out a region of the diagram that is parallel to the time axis; a moving object occupies a slanted region.

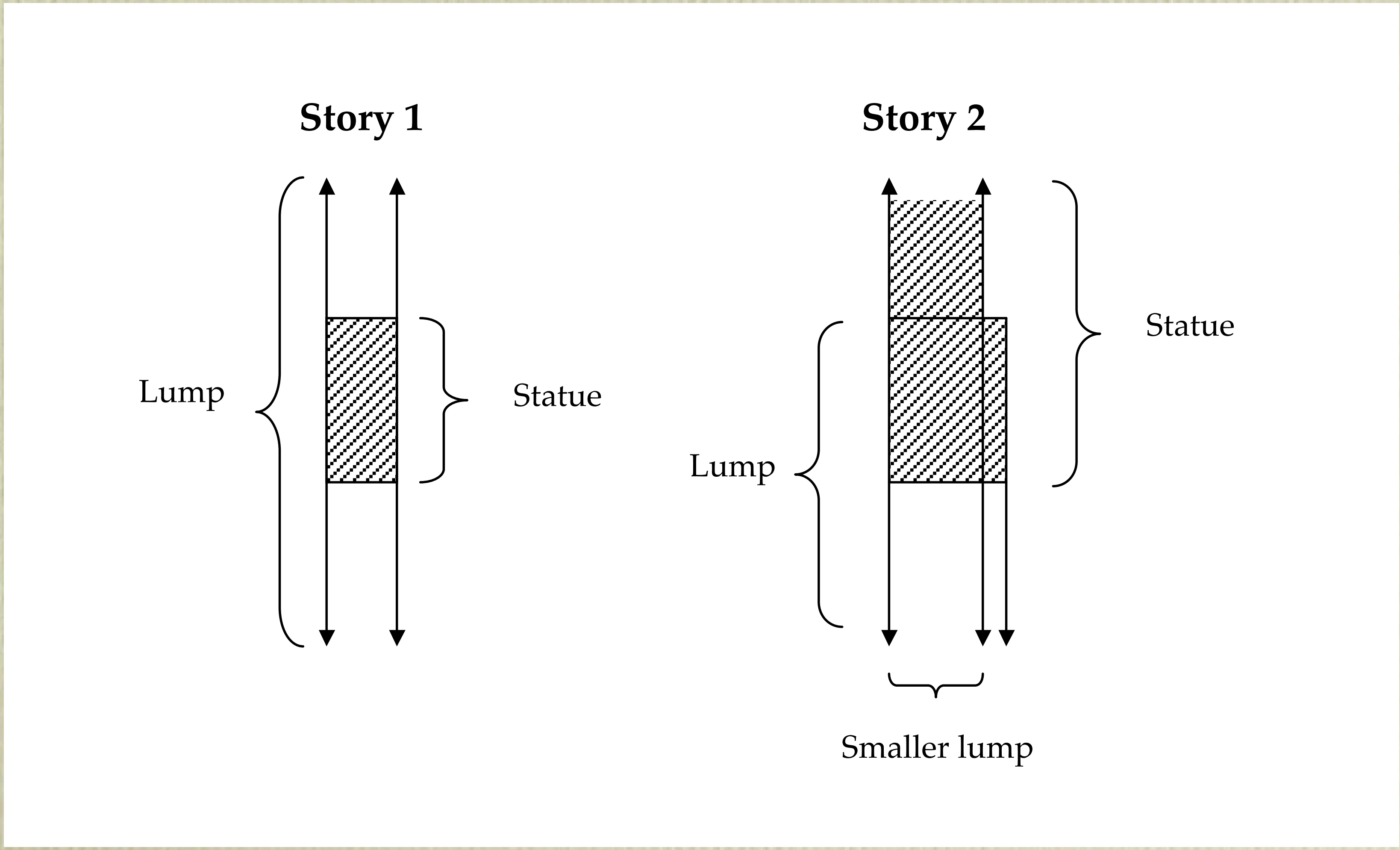


- An ‘ersatz spacetime point’ can be taken to be an ordered pair whose first member is a point of space and whose second member is a point of time.

- Let’s say that an object *traces out* a certain set S of ersatz spacetime points if and only if S is the set of all $<p,t>$ such that p lies within the boundary of the object at t.

Diagrams of statues and lumps

(as drawn by a friend of coincidence)



- Another way to state the principle that two objects can never be in the same place at the same time: it never happens that two objects trace out spacetime regions that have the same intersection with some time.

The analogy between space and time

- Putting the 'one object to a place' principle in this way *might* make it look less compelling.
- Compare: the I-70 and the Pennsylvania Turnpike. The regions of space these two objects occupy are not the same: but their intersection with the state of Pennsylvania is the same.
    - You might say that they coincide *in Pennsylvania*; but it would be silly to think that there can be no two objects that coincide *in a state*.
    - Maybe we can think of times in the same way?

The doctrine of temporal parts

- What Sider calls ‘four-dimensionalism’, though I don’t think that’s a good name.
- Four-dimensionalists typically believe in a *temporally unrelativised notion of parthood* — a way of understanding the claim that *x* is part of *y* on which this isn’t the sort of claim that could be true at one time but not at another.

- Using this notion, the doctrine of temporal parts can be stated as follows:
    - Whenever an object exists at a time, the object has a temporal part at that time.
    - x is a temporal part of y at t means: x exists only at t, and x is part of y, and x shares a part with every part of y that exists at t.

- Sider claims that the doctrine of temporal parts explains how it is possible for two things to be in the same place at the same time:
    - 'At any given time it is only a temporal part of a spacetime worm that is wholly present. Thus it is only temporal parts of Statue and Lump that are wholly present at the time of coincidence. How can these temporal parts both fit into a single region of space? Because 'they' are *identical.*' (6).
    - Q: What does 'wholly present' mean?
    - Q: Is the belief in temporal parts really helping? If I didn't believe that there was such a thing as I-70-in-Allegheny-County, would that make it mysterious how I-70 and the Turnpike could coincide here?